Power System Study Guide

From Classical Network Analysis to Inverter-Based Resources, Grid-Forming Control, and Microgrid Fault Ride-Through

Generated Study Notes

July 13, 2026

Abstract

This guide summarizes core and advanced power-system topics: per-unit normalization, bus modeling, power-flow equations, voltage and angle stability, faults, protection, synchronous-machine dynamics, inverter-based resources (IBRs), grid-following (GFL) and grid-forming (GFM) controls, the stability impact of high IBR penetration, and practical ride-through concepts for microgrids during grid faults. Equations and figures are included to make the material useful as both a review sheet and an implementation reference.

Contents

1 Big Picture

An electric power system converts mechanical, chemical, solar, wind, or stored energy into electric power, transports it through transmission and distribution networks, and delivers it to loads with acceptable voltage, frequency, reliability, power quality, and safety. The traditional grid is dominated by synchronous generators. Modern grids increasingly include inverter-based resources such as wind, solar photovoltaic (PV), battery energy storage systems (BESS), and power-electronic loads.

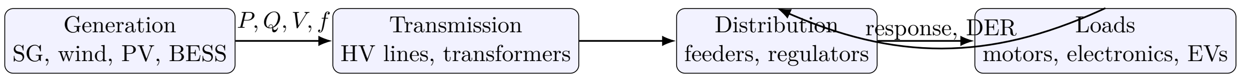

Figure 1: A power system is a controlled energy-conversion network.

2 Phasors, Complex Power, and Per-Unit

2.1 Sinusoidal Steady State

For a sinusoidal voltage,

$$v(t) = \sqrt{2}V_{\mathrm{rms}}\cos(\omega t + \theta), \tag{1}$$

the phasor is

$$\tilde{V} = V_{\mathrm{rms}}\angle\theta. \tag{2}$$

For balanced three-phase systems, a single-phase equivalent is commonly used. Line-to-line and phase quantities obey

$$V_{LL} = \sqrt{3}V_{\phi}, \qquad S_{3\phi} = \sqrt{3}V_{LL}I_L. \tag{3}$$

2.2 Complex Power

Complex power is

$$\tilde{S} = P + \mathrm{j}Q = \tilde{V}\tilde{I}^*, \tag{4}$$

where P is real power and Q is reactive power. The power factor is

$$\mathrm{pf} = \cos\phi = \frac{P}{|\tilde{S}|}. \tag{5}$$

For inductive loads, current lags voltage and $Q > 0$ under the usual load sign convention.

2.3 Per-Unit System

Per-unit quantities reduce transformer-ratio bookkeeping and put impedances on a comparable scale:

$$X_{\mathrm{pu}} = \frac{X_{\mathrm{actual}}}{X_{\mathrm{base}}}. \tag{6}$$

For a three-phase base,

$$Z_{\mathrm{base}} = \frac{V_{LL,\mathrm{base}}^2}{S_{3\phi,\mathrm{base}}}, \qquad I_{\mathrm{base}} = \frac{S_{3\phi,\mathrm{base}}}{\sqrt{3}V_{LL,\mathrm{base}}}. \tag{7}$$

Changing base:

$$Z_{\mathrm{pu,new}} = Z_{\mathrm{pu,old}}\left(\frac{S_{\mathrm{base,new}}}{S_{\mathrm{base,old}}}\right)\left(\frac{V_{\mathrm{base,old}}}{V_{\mathrm{base,new}}}\right)^2. \tag{8}$$

3 Network Modeling

3.1 Transmission-Line Models

For a short line, shunt capacitance can often be neglected:

$$\tilde{V}_S = \tilde{V}_R + \tilde{I}(R + \mathrm{j}X). \tag{9}$$

For medium and long lines, the nominal-π model includes shunt admittance $Y/2$ at both ends.

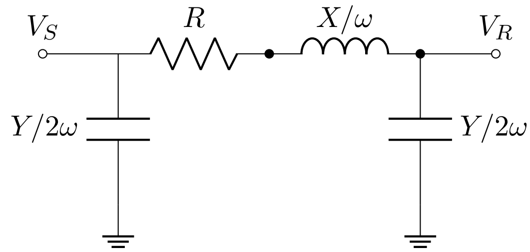

Figure 2: Nominal-π transmission-line model.

3.2 Admittance Matrix

Nodal current injection satisfies

$$\mathbf{I} = \mathbf{Y}_{\text{bus}}\mathbf{V}. \tag{10}$$

For a branch between buses i and k with series admittance y_{ik}:

$$Y_{ii} \leftarrow Y_{ii} + y_{ik} + y_{\text{sh},i}, \tag{11}$$

$$Y_{kk} \leftarrow Y_{kk} + y_{ik} + y_{\text{sh},k}, \tag{12}$$

$$Y_{ik} = Y_{ki} \leftarrow -y_{ik}. \tag{13}$$

Transformer off-nominal tap ratios and phase shifters modify these entries and must be modeled carefully in power-flow and short-circuit studies.

4 Bus Types: Slack, PV, and PQ

Power-flow analysis classifies buses by which variables are specified and which are solved.

Bus type	Known quantities	Unknown quantities	Typical device
Slack / swing	$\lvert V\rvert, \delta$	P, Q	Reference generator
PV / generator	$P, \lvert V\rvert$	Q, δ	Generator with AVR
PQ / load	P, Q	$\lvert V\rvert, \delta$	Load or fixed-power DER

Table 1: Power-flow bus types.

- **Slack bus:** supplies the real and reactive mismatch after all other injections and losses are determined. It sets the system angle reference.
- **PV bus:** real power and voltage magnitude are specified. Reactive output is solved, but if it violates limits, the bus is typically converted to PQ with Q fixed at the limit.
- **PQ bus:** real and reactive powers are specified. Voltage magnitude and angle are solved.

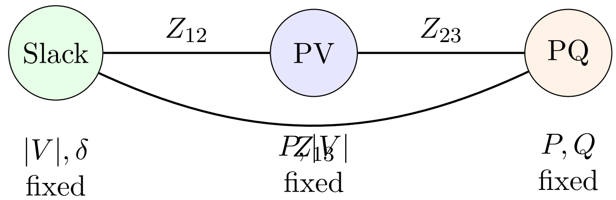

Figure 3: Three classical power-flow bus types.

5 AC Power Flow

5.1 Power Injection Equations

Let $V_i = |V_i|\angle\delta_i$ and $Y_{ik} = G_{ik} + \mathrm{j}B_{ik}$. The complex power injected at bus i is

$$S_i = P_i + \mathrm{j}Q_i = V_i I_i^* = V_i \left(\sum_{k=1}^{n} Y_{ik} V_k \right)^* . \tag{14}$$

The real and reactive power equations are

$$P_i = |V_i| \sum_{k=1}^{n} |V_k| \left(G_{ik} \cos\delta_{ik} + B_{ik} \sin\delta_{ik} \right), \tag{15}$$

$$Q_i = |V_i| \sum_{k=1}^{n} |V_k| \left(G_{ik} \sin\delta_{ik} - B_{ik} \cos\delta_{ik} \right), \tag{16}$$

where $\delta_{ik} = \delta_i - \delta_k$.

5.2 Newton–Raphson Method

Define mismatch vector

$$\Delta x = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P_{\text{spec}} - P_{\text{calc}} \\ Q_{\text{spec}} - Q_{\text{calc}} \end{bmatrix}. \tag{17}$$

At iteration r,

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}^{(r)} = \begin{bmatrix} H & N \\ M & L \end{bmatrix}^{(r)} \begin{bmatrix} \Delta\delta \\ \Delta|V| \end{bmatrix}^{(r)}. \tag{18}$$

Update angles and magnitudes until the maximum mismatch is below tolerance.

5.3 Fast Decoupled Power Flow

For high-voltage transmission systems, $X \gg R$, voltage magnitudes are near 1 pu, and angle differences are small. Then active power is strongly coupled to angle and reactive power to voltage:

$$\Delta P/|V| \approx B'\Delta\delta, \tag{19}$$

$$\Delta Q/|V| \approx B''\Delta|V|. \tag{20}$$

6 Economic Dispatch and Optimal Power Flow

6.1 Economic Dispatch

For generator cost

$$C_i(P_i) = a_i P_i^2 + b_i P_i + c_i, \tag{21}$$

economic dispatch without losses solves

$$\min_{P_i} \sum_i C_i(P_i) \quad \text{s.t.} \quad \sum_i P_i = P_D. \tag{22}$$

The optimality condition is

$$\frac{\mathrm{d}C_i}{\mathrm{d}P_i} = 2a_i P_i + b_i = \lambda, \tag{23}$$

subject to generator limits.

6.2 AC Optimal Power Flow

AC OPF extends power flow by optimizing an objective subject to network constraints:

$$\min_{V,\delta,P_G,Q_G} \quad f(P_G, Q_G, V) \tag{24}$$

$$\text{s.t.} \quad g(V, \delta, P_G, Q_G) = 0, \tag{25}$$

$$h(V, \delta, P_G, Q_G) \leq 0. \tag{26}$$

Constraints include bus voltage limits, branch thermal limits, generator capability curves, reserve requirements, and security constraints.

7 Short-Circuit and Fault Analysis

7.1 Symmetrical Components

Unbalanced three-phase quantities can be decomposed into positive, negative, and zero sequences:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}, \qquad a = e^{\mathrm{j}2\pi/3}. \tag{27}$$

The inverse transform is

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}. \tag{28}$$

7.2 Three-Phase Fault

For a bolted balanced three-phase fault at a bus with Thevenin voltage V_{th} and Thevenin impedance Z_1,

$$I_f = \frac{V_{\mathrm{th}}}{Z_1 + Z_f}. \tag{29}$$

7.3 Single-Line-to-Ground Fault

For an a-phase-to-ground fault:

$$I_a = 3I_0 = \frac{3V_{\mathrm{th}}}{Z_1 + Z_2 + Z_0 + 3Z_f}. \tag{30}$$

Grounding and transformer connections strongly affect Z_0 and therefore ground-fault current.

7.4 Fault Current From Synchronous Machines and IBRs

Synchronous generators can contribute large transient currents, often several per-unit initially, limited by subtransient reactance:

$$I_{\mathrm{sc,SG}} \approx \frac{E''}{X''_d}. \tag{31}$$

IBRs are current-limited by semiconductors and controls, commonly around 1.1–2.0 pu depending on design and grid code:

$$|\tilde{I}_{\mathrm{IBR}}| \leq I_{\mathrm{max}}. \tag{32}$$

This changes protection sensitivity, directionality, relay coordination, and fault-location methods.

8 Protection Fundamentals

Protection detects abnormal conditions and isolates faulted equipment quickly. Common functions include:

- **50/51 overcurrent:** instantaneous/time-delayed current pickup.
- **67 directional overcurrent:** uses voltage polarization to infer fault direction.
- **21 distance:** estimates apparent impedance $Z_{\mathrm{app}} = V/I$.
- **87 differential:** compares currents entering and leaving a zone.
- **27/59 undervoltage/overvoltage** and **81U/O frequency** functions.
- **ROCOF:** rate-of-change-of-frequency, $\mathrm{d}f/\mathrm{d}t$.

IBR Protection Challenge

High IBR penetration reduces short-circuit current magnitude, changes current phase angle through control actions, and can make conventional overcurrent protection less dependable. Protection may need adaptive settings, negative- sequence quantities, differential schemes, transfer trip, traveling waves, or communications-assisted logic.

9 Voltage Control and Reactive Power

9.1 Reactive Power and Voltage

In transmission networks with $X \gg R$, reactive power strongly affects voltage. A useful approximate relation is

$$\Delta V \approx \frac{XQ + RP}{V}. \tag{33}$$

Devices for voltage support include generator automatic voltage regulators (AVRs), shunt capacitors/reactors, static VAR compensators (SVC), STATCOMs, load tap changers, and inverter volt-var controls.

9.2 Generator Capability

Synchronous-generator reactive power is limited by armature current, field current, end-region heating, and stability. IBR reactive capability is limited by current and dc-link constraints:

$$P^2 + Q^2 \le S_{\text{rated}}^2, \qquad I_d^2 + I_q^2 \le I_{\max}^2. \tag{34}$$

During faults, priority may be given to reactive current injection:

$$I_q^* = k_{\text{FRT}} \left(V_{\text{ref}} - V_{\text{PCC}} \right), \qquad |I^*| \le I_{\max}. \tag{35}$$

10 Frequency Control and Inertia

10.1 Swing Equation

The synchronous-machine rotor angle follows

$$M \frac{\mathrm{d}^2 \delta}{\mathrm{d}t^2} = P_m - P_e - D \frac{\mathrm{d}\delta}{\mathrm{d}t}, \tag{36}$$

or, in frequency form,

$$\frac{2H}{f_0} \frac{\mathrm{d}f}{\mathrm{d}t} = P_m - P_e - D_f (f - f_0), \tag{37}$$

where H is the inertia constant. Immediately after a generation-loss event, frequency rate of change is approximately

$$\frac{\mathrm{d}f}{\mathrm{d}t} \approx \frac{f_0}{2H_{\text{sys}}} \left(\frac{\Delta P}{S_{\text{base}}} \right). \tag{38}$$

10.2 Primary, Secondary, and Tertiary Control

- **Primary control:** governor or inverter droop responds in seconds.
- **Secondary control:** automatic generation control restores frequency and tie-line interchange in tens of seconds to minutes.
- **Tertiary control:** economic redispatch and reserve management.

11 Stability Concepts

11.1 Rotor-Angle Stability

Rotor-angle stability is the ability of synchronous machines to remain in synchronism. For a classical generator connected to an infinite bus:

$$P_e = \frac{EV}{X} \sin \delta. \tag{39}$$

The synchronizing torque coefficient is

$$K_s = \frac{\partial P_e}{\partial \delta} = \frac{EV}{X} \cos \delta. \tag{40}$$

As δ approaches 90°, synchronizing torque decreases.

11.2 Equal-Area Criterion

For a single-machine infinite-bus system, transient stability can be assessed by comparing accelerating and decelerating areas on the P–δ curve:

$$A_{\mathrm{acc}} = \int_{\delta_0}^{\delta_c} (P_m - P_{e,\mathrm{fault}})\,\mathrm{d}\delta, \qquad A_{\mathrm{dec}} = \int_{\delta_c}^{\delta_{\max}} (P_{e,\mathrm{post}} - P_m)\,\mathrm{d}\delta. \tag{41}$$

Stability requires $A_{\mathrm{dec}} \geq A_{\mathrm{acc}}$.

11.3 Voltage Stability

Voltage stability is the ability to maintain acceptable voltage after a disturbance. A simple radial two-bus example shows the "nose curve":

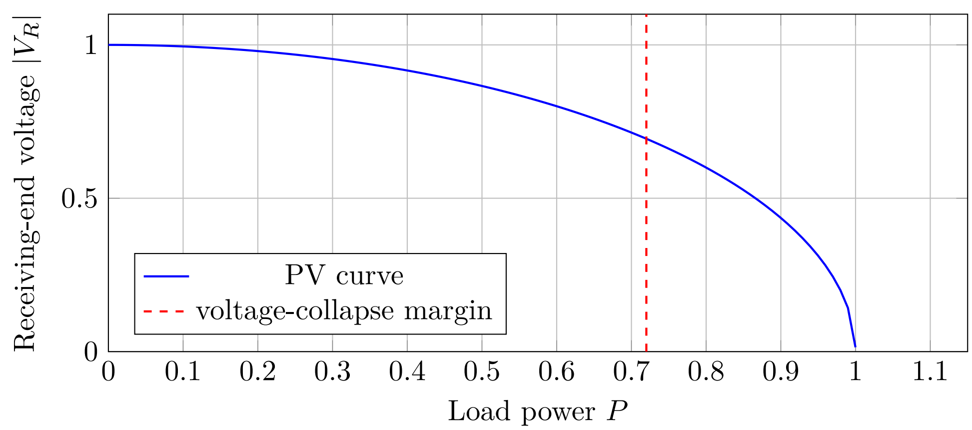

Figure 4: Conceptual PV curve showing maximum loadability and voltage-collapse margin.

11.4 Small-Signal Stability

Linearize nonlinear dynamics $\dot{x} = f(x, u)$ around an operating point:

$$\Delta\dot{x} = A\Delta x + B\Delta u. \tag{42}$$

Eigenvalues $\lambda_i = \sigma_i + \mathrm{j}\omega_i$ determine modal behavior. The damping ratio is

$$\zeta_i = \frac{-\sigma_i}{\sqrt{\sigma_i^2 + \omega_i^2}}. \tag{43}$$

Negative damping or weak inter-area modes can be worsened by controls that interact with grid impedance, PLLs, or other converters.

12 Power Quality

Power quality includes voltage sags, swells, interruptions, harmonics, flicker, unbalance, and transients. Harmonic distortion is quantified by

$$\mathrm{THD}_V = \frac{\sqrt{V_2^2 + V_3^2 + \cdots + V_n^2}}{V_1}. \tag{44}$$

Converters use filters and controls to meet harmonic limits. Resonance can occur when inverter filters, cable capacitance, transformer leakage, and grid impedance form lightly damped modes.

13 Inverter-Based Resources

13.1 What Is an IBR?

An IBR interfaces an energy source or storage device through power electronics. Examples include PV inverters, wind turbine full converters, type-3 wind generators with partial converters, BESS, fuel cells, and many HVDC terminals.

13.2 Converter Power Equations in dq Coordinates

With the d-axis aligned to voltage, $v_q = 0$:

$$P = \frac{3}{2}(v_d i_d + v_q i_q) = \frac{3}{2} v_d i_d, \tag{45}$$

$$Q = \frac{3}{2}(v_q i_d - v_d i_q) = -\frac{3}{2} v_d i_q. \tag{46}$$

Thus i_d controls active power and i_q controls reactive power under this axis convention.

13.3 Grid-Following (GFL) Inverters

A GFL inverter behaves like a controlled current source. It synchronizes to an existing grid voltage using a phase-locked loop (PLL), then injects commanded currents.

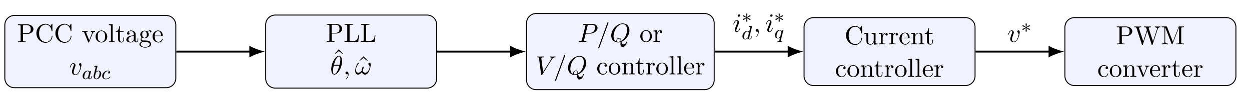

Figure 5: Grid-following inverter control is synchronized by the measured grid voltage.

Typical GFL PLL dynamics include

$$\dot{\hat{\theta}} = \omega_0 + K_p v_q + K_i \int v_q \, \mathrm{d}t. \tag{47}$$

GFL inverters are effective in strong grids but can struggle in weak grids or islanded microgrids because they require a voltage reference to follow.

13.4 Grid-Forming (GFM) Inverters

A GFM inverter behaves more like a controllable voltage source behind an impedance. It establishes voltage magnitude and frequency rather than relying entirely on an external voltage.

Common GFM approaches:

- **Droop control:**

$$\omega = \omega_0 - m_p(P - P_0), \tag{48}$$

$$V = V_0 - n_q(Q - Q_0). \tag{49}$$

- **Virtual synchronous machine (VSM):**

$$2H_\mathrm{v}\dot{\omega} = P^* - P - D_\mathrm{v}(\omega - \omega_0). \tag{50}$$

- **Dispatchable virtual oscillator control:** nonlinear oscillator dynamics regulate voltage waveform and share power.

- **Matching control:** uses dc-link dynamics to emulate synchronous-machine electromechanical coupling.

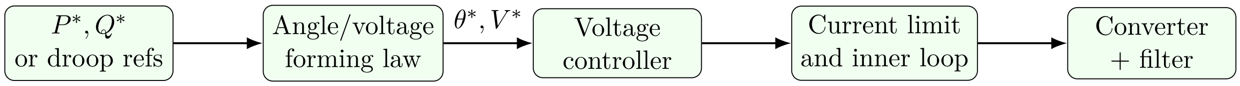

Figure 6: Grid-forming inverter control creates a voltage/frequency reference.

13.5 GFL vs GFM Summary

Feature	Grid-following	Grid-forming
Equivalent behavior	Current source	Voltage source behind impedance
Synchronization	PLL follows grid voltage	Internal oscillator or swing-like law
Islanded operation	Not sufficient alone	Can support islanded microgrid
Weak-grid performance	PLL/control interactions possible	Can improve voltage/frequency stiffness
Fault response	Current-limited injection	Must handle current limiting without losing voltage-forming behavior
Power sharing	Usually outer-loop command	Droop/VSM sharing natural

14 How IBRs Affect Power-Grid Stability

14.1 Reduced Inertia and Faster Dynamics

Replacing synchronous machines with IBRs reduces physical rotating inertia unless the IBR control intentionally provides synthetic inertia or fast frequency response. Lower inertia produces higher ROCOF:

$$\left|\frac{\mathrm{d}f}{\mathrm{d}t}\right| \propto \frac{|\Delta P|}{H_{\mathrm{sys}}}. \tag{51}$$

However, IBRs can respond much faster than mechanical governors if controls, energy headroom, and communications are designed correctly.

14.2 PLL and Weak-Grid Instability

Grid strength is often measured using short-circuit ratio:

$$\mathrm{SCR} = \frac{S_{\mathrm{sc}}}{S_{\mathrm{IBR,rated}}}, \qquad S_{\mathrm{sc}} = \frac{V^2}{|Z_{\mathrm{th}}|}. \tag{52}$$

Low SCR means the inverter's own current injection significantly moves the PCC voltage. For GFL controls, PLL dynamics can couple with the weak grid and create oscillations. Effective grid strength can also be reduced by nearby converters interacting through the same network.

14.3 Current Limiting and Protection

IBRs cannot supply unlimited fault current. During severe voltage depressions, the current limit

$$\sqrt{i_d^2 + i_q^2} \le I_{\max} \tag{53}$$

forces a choice between active current, reactive support, negative-sequence current, and dc-link protection. This affects voltage recovery, relay dependability, and transient stability.

14.4 Control Interactions

High IBR penetration introduces many fast controls: PLLs, current loops, dc-link loops, volt-var controls, plant-level controllers, and grid-forming controllers. Interactions can occur over milliseconds to seconds. Stability studies must include electromagnetic transient (EMT) simulations for fast converter controls when RMS phasor models are insufficient.

14.5 Fault-Induced Delayed Voltage Recovery

IBRs often ride through voltage sags and inject reactive current, but motor loads, feeder controls, and current-limited converters may cause slow voltage recovery. Coordination among DER ride-through, distribution protection, and transmission voltage controls is essential.

15 Microgrids

15.1 Definition and Modes

A microgrid is a local system of loads, DERs, storage, protection, and controls that can operate grid-connected or islanded. It needs:

- at least one grid-forming source for islanded voltage/frequency,
- energy management for power balance,
- protection that works for both grid-connected and islanded fault levels,
- black-start and resynchronization procedures.

15.2 Microgrid Control Hierarchy

1. **Primary control:** local voltage/frequency regulation, droop, current limiting, and virtual impedance.
2. **Secondary control:** restores frequency/voltage deviations caused by droop and coordinates multiple DERs.
3. **Tertiary control:** economic dispatch and power exchange with the main grid.

16 Ride-Through During Faults

16.1 Meaning of Ride-Through

Ride-through means that DERs remain connected and support the grid or microgrid during abnormal voltage/frequency events instead of tripping immediately. Requirements are usually defined by voltage–time and frequency–time envelopes.

16.2 Core Ride-Through Mechanisms

Fault ride-through for an inverter-dominated microgrid usually combines:

1. **Fast fault detection:** voltage sag, current rise, negative/zero sequence, impedance, or differential signals.

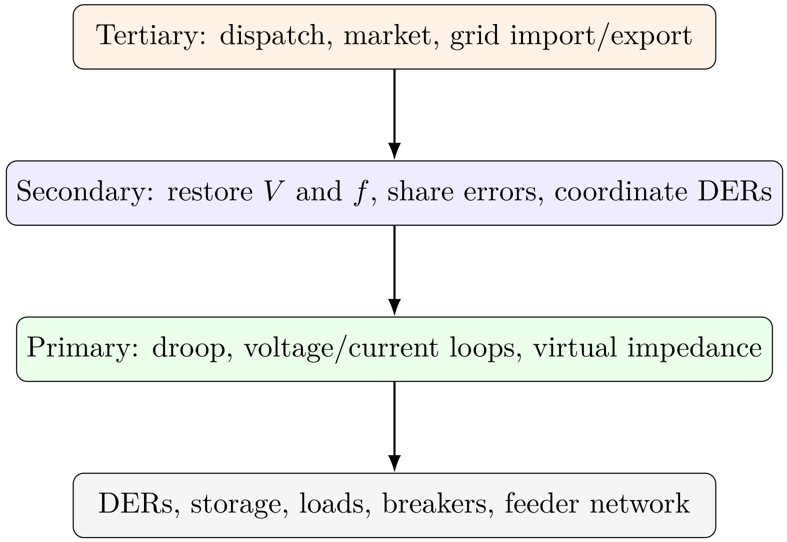

Figure 7: Microgrid hierarchical control.

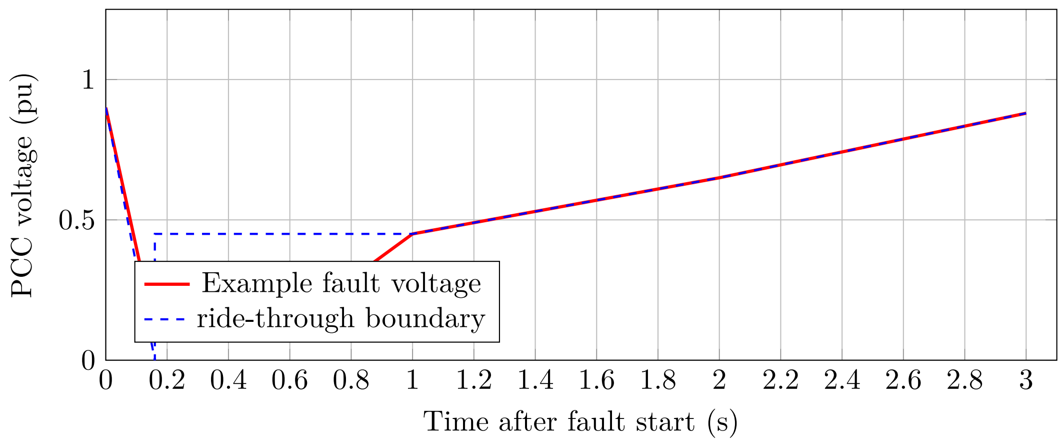

Figure 8: Conceptual low-voltage ride-through envelope. Actual limits depend on grid code.

2. **Current limiting:** protects semiconductors and filters while maintaining a controlled current vector.

3. **Reactive current priority:** supports voltage at the PCC during low-voltage events.

4. **Active power reduction:** prevents dc-link overvoltage when ac power export is limited by sagged voltage.

5. **Energy buffering:** BESS, dc-link capacitance, braking chopper, or curtailment absorbs mismatch.

6. **Protection coordination:** isolates only the faulted zone.

7. **Post-fault recovery:** ramps current, voltage, and power references to avoid secondary dips or converter interactions.

16.3 Current-Limited Control During a Fault

Let $I_{\max}$ be the converter current limit. A common current-priority logic is

$$i_q^* = \mathrm{sat}\left(k_v(V_{\mathrm{ref}} - V_{\mathrm{PCC}}), -I_{\max}, I_{\max}\right), \tag{54}$$

$$i_d^* = \mathrm{sgn}(P^*)\sqrt{\max\left(I_{\max}^2 - (i_q^*)^2, 0\right)}. \tag{55}$$

If the inverter uses the opposite dq sign convention, the sign of i_q must be changed. For GFM inverters, the current limit must be integrated without abruptly collapsing the voltage reference. Common methods are virtual impedance, voltage reference saturation, current reference limiting, and mode switching to a protected current-source behavior.

16.4 GFM Fault Ride-Through

GFM ride-through is difficult because a voltage source behind low impedance would naturally produce very high fault current. Practical GFM controls add a virtual impedance:

$$v_{\mathrm{cmd}} = v_{\mathrm{gfm}} - R_v i - L_v \frac{\mathrm{d}i}{\mathrm{d}t}, \tag{56}$$

or reduce voltage reference when current approaches the limit:

$$V_{\mathrm{lim}}^* = V^* \min\left(1, \frac{I_{\max}}{|I|}\right). \tag{57}$$

After clearing, the controller restores normal voltage-forming behavior with rate limits:

$$\left|\frac{\mathrm{d}P^*}{\mathrm{d}t}\right| \le r_P, \qquad \left|\frac{\mathrm{d}V^*}{\mathrm{d}t}\right| \le r_V. \tag{58}$$

16.5 Microgrid Fault-Ride-Through Sequence

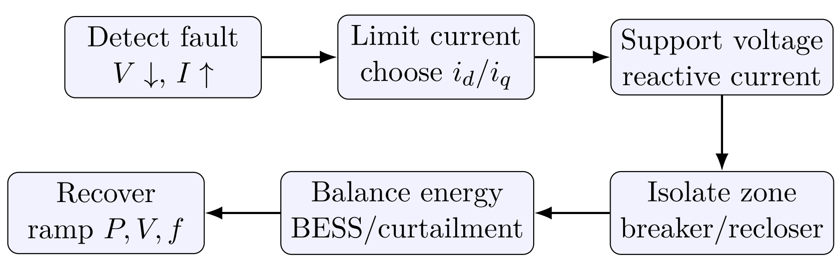

Figure 9: A practical sequence for microgrid fault ride-through.

16.6 Islanded Microgrid Faults

Islanded microgrids have much lower fault current than utility-connected systems, especially when dominated by IBRs. Protection may use:

- differential protection inside critical zones,
- directional elements with adaptive settings,
- voltage-based and sequence-based detection,

- communication-assisted trip,
- temporary overcurrent capability from selected grid-forming BESS,
- fast solid-state breakers or hybrid breakers.

For unbalanced faults, GFM controls may regulate positive sequence while selectively injecting negative-sequence current:

$$\begin{bmatrix} i_1^* \\ i_2^* \end{bmatrix} = \begin{bmatrix} f_1(V_1, P^*, Q^*) \\ f_2(V_2, I_{\max}) \end{bmatrix}, \qquad |i_1^* + i_2^*| \leq I_{\max}. \tag{59}$$

17 Advanced Study Methods

17.1 RMS Phasor vs EMT Simulation

- **RMS/transient-stability simulation:** efficient for electromechanical dynamics, governor/AVR response, and large systems.
- **EMT simulation:** necessary for converter switching, PLL dynamics, unbalanced faults, harmonics, sub-synchronous/control interactions, and protection details.

High IBR systems often require EMT model validation because average phasor models may hide fast control instabilities.

17.2 Impedance-Based Stability

Converter-grid interactions can be studied using source and load impedances. A small-signal stability condition can be framed with the minor-loop gain

$$L(s) = Z_{\text{source}}(s) Y_{\text{load}}(s), \tag{60}$$

and Nyquist criteria. Weak grids have larger Z_{source}, making interaction with converter input admittance more likely.

17.3 Resonance and Sub-Synchronous Oscillation

Series compensation, wind plants, HVDC, and converter controls can create oscillations below or above the fundamental frequency. Modal damping, impedance scans, and EMT studies are used to identify risk.

18 Common Exam and Interview Questions

1. Why is one bus chosen as slack in power flow?
2. What happens when a PV bus reaches reactive power limits?
3. Why does P mostly depend on angle and Q mostly on voltage in a transmission grid?
4. How do IBR fault currents differ from synchronous-generator fault currents?
5. Why can a PLL destabilize a GFL inverter in a weak grid?
6. What makes a GFM inverter useful for an islanded microgrid?

7. How is low-voltage ride-through achieved without damaging the converter?
8. Why can high IBR penetration require EMT instead of only RMS simulation?
9. How do virtual impedance and current limiting help GFM fault ride-through?
10. What protection functions remain dependable when fault current is low?

19 Quick Reference

Essential Equations

$$
\begin{aligned}
S &= VI^* = P + \mathrm{j}Q \\
Z_{\text{base}} &= V_{\text{base}}^2 / S_{\text{base}} \\
I &= Y_{\text{bus}} V \\
P_i &= |V_i| \sum_k |V_k| (G_{ik} \cos\delta_{ik} + B_{ik} \sin\delta_{ik}) \\
Q_i &= |V_i| \sum_k |V_k| (G_{ik} \sin\delta_{ik} - B_{ik} \cos\delta_{ik}) \\
P_e &= \frac{EV}{X} \sin\delta \\
2H\dot{\omega} &= P_m - P_e - D(\omega - \omega_0) \\
\text{SCR} &= S_{\text{sc}} / S_{\text{IBR,rated}} \\
I_{\text{IBR}} &\leq I_{\max}
\end{aligned}
$$

Study Priorities

- Know bus types and power-flow equations.
- Understand voltage, angle, and frequency stability separately.
- Compare SG and IBR fault behavior.
- Distinguish GFL current-source behavior from GFM voltage-source behavior.
- For microgrids, connect controls, energy balance, and protection.

20 Suggested Learning Path

1. Review phasors, complex power, and per-unit calculations.
2. Build Y_{bus} for a three-bus system and solve power flow.
3. Study symmetrical components and compute SLG and three-phase faults.
4. Simulate swing-equation response to a generation trip.
5. Compare GFL and GFM inverter block diagrams and control equations.
6. Analyze how current limits alter fault response and voltage support.
7. Study a microgrid fault sequence: detect, limit, support, isolate, recover.
8. Move from RMS to EMT simulation when converter controls dominate dynamics.